PowerBI - HR Dashboard

Data Analysis Report: Insights from Power BI HR Dashboard

Project Overview

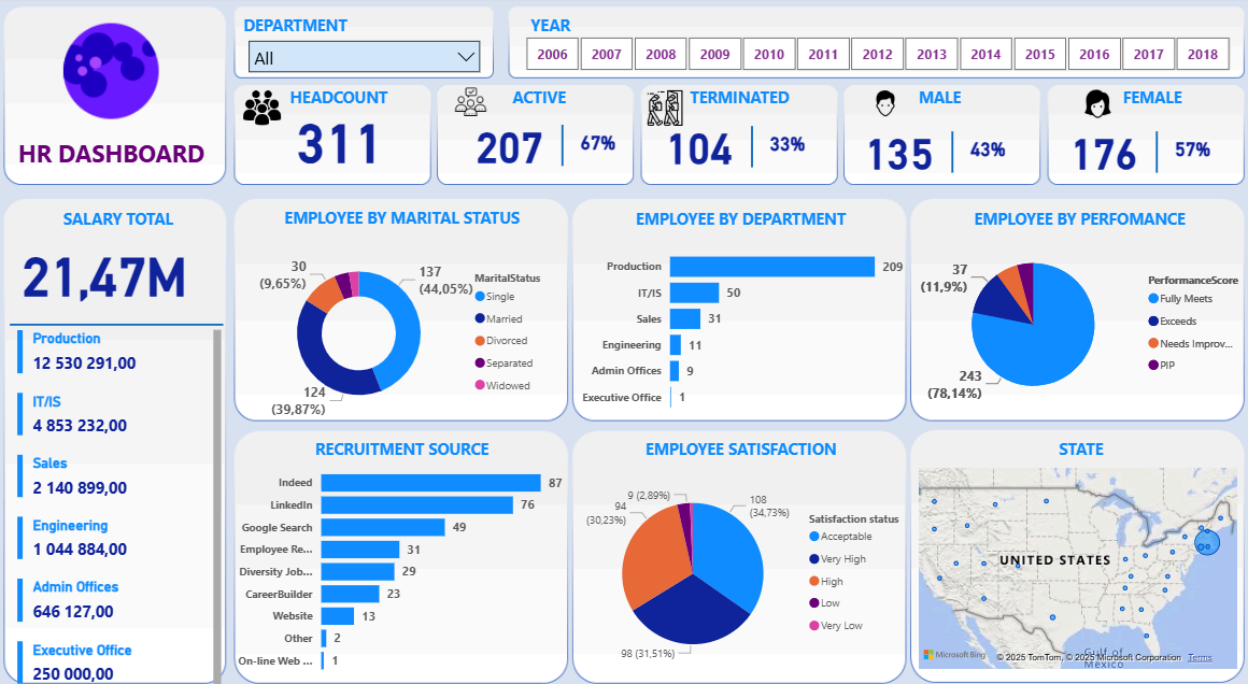
This project involves a comprehensive analysis of key HR metrics using a Power BI dashboard. The dashboard was developed to provide actionable insights into workforce demographics, employee performance, satisfaction, and key trends, which are crucial for strategic human resource management and decision-making.

Methodology

The analysis in this report is based on a Power BI dashboard created using these data preparation steps and measures:

A new column was added in Power Query to group employees into two categories: "Active" and "Terminated." Another column was added to categorize employees by their "Satisfaction Status" based on their satisfaction score. Several measures were created to calculate key metrics, including:

- Active Employees and Active %
- Female Employees and Female %
- Headcount
- Male Employees and Male %
- Salary
- Terminated Employees and Terminated %



Overall Workforce Analysis

Key Metrics and Analysis:

This dashboard provides a snapshot of the company's workforce. Key metrics include a headcount of 311, with 207 (67%) active employees and 104 (33%) terminated employees. The gender breakdown shows a female majority, with 176 (57%) female employees and 135 (43%) male employees. The dashboard also indicates total salary spending of $21.47 million.

Trends and Patterns:

- Termination Rate: The 33% termination rate is significant and requires further investigation. This figure indicates considerable turnover within the company.
- Gender Distribution: The workforce has a female majority, which is important for understanding workforce composition and potential equity considerations.

- Recruitment Sources: The dashboard reveals that the top three recruitment sources are Indeed (87), LinkedIn (76), and Google Search (49), highlighting the most effective channels for talent acquisition.

Implications:

The high termination rate is a primary concern that needs addressing. Understanding why employees leave whether due to low satisfaction or poor performance is vital for improving retention. The effectiveness of online recruitment platforms like Indeed and LinkedIn suggests these channels should be prioritized in future hiring strategies.

Employee Performance and Satisfaction Analysis
Key Metrics and Analysis:
This section examines the performance and satisfaction of the current workforce, focusing on the distribution of employees by performance score and satisfaction status.

Trends and Patterns:

- Performance Distribution: Most employees are high performers, with 243 (78.14%) exceeding expectations ("Fully Meets," "Exceeds," or "Needs Improvement"). Only a small percentage of employees are underperforming.
- Employee Satisfaction: Satisfaction data presents a mixed picture. The largest group falls in the "Acceptable" category with 108 (34.73%) employees, followed by "Very High" with 96 (31.51%) employees. However, a significant number of employees report "Low" or "Very Low" satisfaction, potentially contributing to the high termination rate.

Implications:

The gap between high performance scores and mixed satisfaction levels is a critical finding. It suggests that while employees perform well, many may be unhappy. This could indicate issues with work-life balance, compensation, or career development opportunities, potentially leading to voluntary departures. Comparing satisfaction data with termination data could reveal whether low satisfaction predicts turnover.

1. **Departmental and Marital Status Analysis**
   Key Metrics and Analysis:

This dashboard also segments employees by department and marital status, offering a deeper look at workforce composition.

Trends and Patterns:

- Departmental Breakdown: The Production department has the largest workforce (209 employees), followed by IT/IS (50) and Sales (31). This distribution is typical, as Production is often the largest department in many companies.
- Marital Status: Most employees are Married (137 or 44.05%) or Single (124 or 39.87%), providing a demographic snapshot of the workforce.

Implications:

With Production housing the majority of employees, any issues in this department such as low satisfaction or high turnover could significantly impact company-wide metrics. The marital status data can help in understanding employees' diverse needs and tailoring benefits and policies accordingly.

Synthesizing Insights Across the Dashboard

The insights from this dashboard interconnect to provide a comprehensive view of the employee lifecycle. The 33% termination rate stands out as a central finding. Despite most employees being high performers, the mixed satisfaction data with many reporting "Low" or "Very Low" satisfaction may be a key factor driving terminations. The concentration of employees in Production means that satisfaction issues in this department could significantly impact the overall termination rate. The recruitment source data offers valuable insights for enhancing hiring efforts, while the departmental and demographic breakdowns can help address specific employee group needs to improve retention.